



Continuum Mechanics

Steven M. Wise

swise1@utk.edu
University of Tennessee

Fall 2026



Chapter 04

Kinematics

Part 1

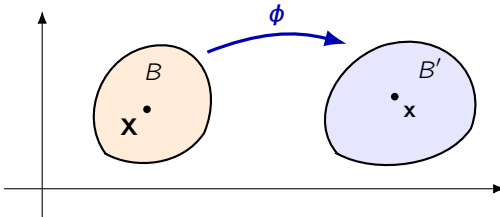


Deformations



Definition (Reference and Deformed Configurations)

Let $B \subseteq \mathbb{E}^3$ be an undeformed configuration of a body, and let $B' \subseteq \mathbb{E}^3$ be its deformed configuration. B is called the *reference configuration*, and B' is called the *deformed configuration*. The two sets describe the same body before and after deformation.





Definition (Material and Spatial Coordinates)

Points of the reference configuration are denoted by

$$\mathbf{X} = (X_1, X_2, X_3)^T \in B.$$

Points of the deformed configuration are denoted by

$$\mathbf{x} = (x_1, x_2, x_3)^T \in B'.$$

The coordinates X_i are called *material coordinates*, while the coordinates x_i are called *spatial coordinates*.



Definition (Deformation Map and Displacement)

The deformation of a body from the reference configuration B into the deformed configuration B' is described by a function

$$\phi : B \longrightarrow B',$$

which maps each material point $\mathbf{X} \in B$ to the spatial point

$$\mathbf{x} = \phi(\mathbf{X}) \in B'.$$

The function ϕ is called the *deformation map* relative to B . The *displacement field* relative to ϕ is defined via

$$\mathbf{u}(\mathbf{X}) = \phi(\mathbf{X}) - \mathbf{X}. \quad (4.1)$$

Thus $\mathbf{u} : B \rightarrow \mathcal{V}$ is a vector field on the reference configuration.



Definition (Admissible Deformation)

Let $B \subseteq \mathbb{E}^3$ be a reference configuration, and suppose

$$\phi : B \longrightarrow B'$$

is its deformation map. The map ϕ is called *admissible* if

① $\phi : B \rightarrow B'$ is one-to-one and onto;

②

$$\det \nabla \phi(\mathbf{X}) > 0 \quad \text{for every } \mathbf{X} \in B.$$

The second condition requires an admissible deformation to preserve the self-orientation of the body. In particular, a body cannot be continuously deformed into its mirror image.



The Deformation Gradient



Definition (Deformation Gradient)

Suppose that $\phi : B \rightarrow B'$ is a smooth admissible deformation map. The *deformation gradient* is the tensor field

$$F : B \longrightarrow \mathcal{V}^2$$

defined by

$$F(\mathbf{X}) = \nabla \phi(\mathbf{X}).$$

The tensor field F characterizes the local behavior of the deformation.

Local Linearization



Taylor's theorem about a point $\mathbf{X}_0 \in B$ gives

$$\phi(\mathbf{X}) = \phi(\mathbf{X}_0) + F(\mathbf{X}_0)(\mathbf{X} - \mathbf{X}_0) + O(|\mathbf{X} - \mathbf{X}_0|^2).$$

Equivalently,

$$\phi(\mathbf{X}) = \mathbf{c} + F(\mathbf{X}_0)\mathbf{X} + O(|\mathbf{X} - \mathbf{X}_0|^2), \quad (4.2)$$

where

$$\mathbf{c} := \phi(\mathbf{X}_0) - F(\mathbf{X}_0)\mathbf{X}_0.$$

Thus $F(\mathbf{X}_0)$ determines the first-order behavior of ϕ near $\mathbf{X}_0$.



Definition (Homogeneous Deformation)

A deformation $\phi : B \rightarrow B'$ is called *homogeneous* if

$$F(\mathbf{X}) = F_0 \quad \text{for every } \mathbf{X} \in B,$$

where $F_0 \in \mathcal{V}^2$ is constant.

In this case, the approximation in (4.2) is exact. Hence

$$\phi(\mathbf{X}) = \mathbf{c} + F\mathbf{X} \quad \text{for every } \mathbf{X} \in B. \quad (4.3)$$



Definition (Translation)

A deformation ϕ is called a *translation* if

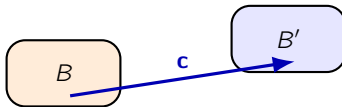
$$\phi(\mathbf{X}) = \mathbf{X} + \mathbf{c}$$

for a constant vector $\mathbf{c} \in \mathcal{V}$.

For a translation,

$$F = I, \quad \mathbf{u}(\mathbf{X}) = \phi(\mathbf{X}) - \mathbf{X} = \mathbf{c} \quad \text{for every } \mathbf{X} \in B.$$

Thus a translation is homogeneous and changes neither the shape nor the orientation of the body.





Definition (Fixed Point)

Suppose $\phi : B \rightarrow B'$ is a homogeneous deformation,

$$\phi(\mathbf{X}) = \mathbf{c} + F\mathbf{X}.$$

The map ϕ is said to have a *fixed point* at $\mathbf{Y} \in B$ if

$$\phi(\mathbf{X}) = \mathbf{Y} + F(\mathbf{X} - \mathbf{Y}) \quad \text{for every } \mathbf{X} \in B. \quad (4.4)$$

Indeed, equation (4.4) gives

$$\phi(\mathbf{Y}) = \mathbf{Y}.$$



Result 4.1

Let ϕ be an arbitrary homogeneous deformation with constant deformation gradient $F \in \mathcal{V}^2$. Given any point $\mathbf{Y}$, the map can be decomposed as

$$\phi = \mathbf{d}_1 \circ \mathbf{g} = \mathbf{g} \circ \mathbf{d}_2,$$

where $\mathbf{d}_1$ and $\mathbf{d}_2$ are translations and $\mathbf{g}$ is homogeneous with the fixed point $\mathbf{Y}$. In particular,

$$\mathbf{g}(\mathbf{X}) = \mathbf{Y} + F(\mathbf{X} - \mathbf{Y}).$$

Proof.

Exercise. □



Rotations and Stretches



Definition (Rotation About a Fixed Point)

A homogeneous deformation ϕ is called a *rotation* about a fixed point $\mathbf{Y}$ if

$$\phi(\mathbf{X}) = \mathbf{Y} + Q(\mathbf{X} - \mathbf{Y}),$$

where Q is a rotation tensor:

$$Q^T Q = I, \quad \det(Q) = 1.$$



Example (Rotation of a Cuboid About an Off-Center Point)

Consider the cuboid

$$B = (0, 1) \times (0, 1) \times (-1, 1)$$

and the fixed point

$$\mathbf{Y} = \left(\frac{1}{4}, \frac{1}{4}, 0 \right)^T.$$

Let us consider a rotation about $\mathbf{Y}$ of the cuboid through

$$\theta = 45^\circ = \frac{\pi}{4}$$

counterclockwise in the X_1 - X_2 plane, as viewed from the positive X_3 -axis. Thus, the rotation axis is the line parallel to the X_3 -axis that passes through $\mathbf{Y}$.

The rotation tensor is

$$Q = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$



Example (Rotation of a Cuboid, Cont.)

The rotation about $\mathbf{Y}$ is

$$\phi(\mathbf{X}) = \mathbf{Y} + Q(\mathbf{X} - \mathbf{Y}).$$

Therefore,

$$\begin{aligned}x_1 &= \frac{1}{4} + \frac{1}{\sqrt{2}} \left[\left(X_1 - \frac{1}{4} \right) - \left(X_2 - \frac{1}{4} \right) \right] = \frac{1}{4} + \frac{X_1 - X_2}{\sqrt{2}}, \\x_2 &= \frac{1}{4} + \frac{1}{\sqrt{2}} \left[\left(X_1 - \frac{1}{4} \right) + \left(X_2 - \frac{1}{4} \right) \right] = \frac{1}{4} + \frac{X_1 + X_2 - \frac{1}{2}}{\sqrt{2}}, \\x_3 &= X_3.\end{aligned}$$

In particular, $\phi(\mathbf{Y}) = \mathbf{Y}$. Also,

$$Q^T Q = I, \quad \det(Q) = 1,$$

so Q is a rotation tensor.



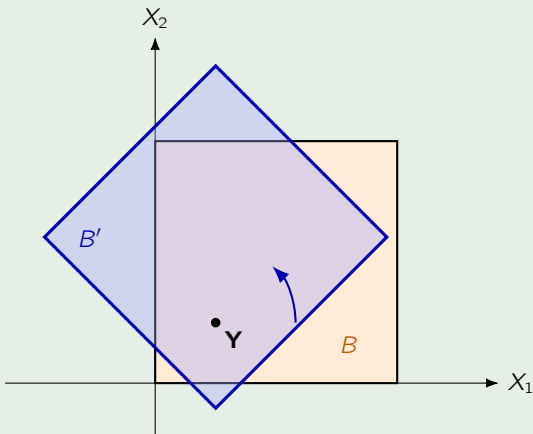
Example (Two-Dimensional View of the Rotation)

The X_1 – X_2 cross-section of B is the unit square. Its four vertices map as follows:

(X_1, X_2)	(x_1, x_2)
$(0, 0)$	$\left(\frac{1}{4}, \frac{1}{4} - \frac{\sqrt{2}}{4}\right)$
$(1, 0)$	$\left(\frac{1}{4} + \frac{\sqrt{2}}{2}, \frac{1}{4} + \frac{\sqrt{2}}{4}\right)$
$(1, 1)$	$\left(\frac{1}{4}, \frac{1}{4} + \frac{3\sqrt{2}}{4}\right)$
$(0, 1)$	$\left(\frac{1}{4} - \frac{\sqrt{2}}{2}, \frac{1}{4} + \frac{\sqrt{2}}{4}\right)$

Since $x_3 = X_3$, every horizontal cross-section undergoes the same planar rotation.

Example (Two-Dimensional View of the Rotation, Cont.)



The orange square is the original cross-section $(0, 1) \times (0, 1)$. The blue square is its image under the rotation.



Definition (Stretch From a Fixed Point)

A homogeneous deformation ϕ is called a *stretch* from a fixed point $\mathbf{Y}$ if

$$\phi(\mathbf{X}) = \mathbf{Y} + S(\mathbf{X} - \mathbf{Y}),$$

where S is a symmetric positive-definite tensor.



Example (A Stretch of a Rectangular Body)

Let

$$\mathbf{Y} = (2, 2, 2)^T$$

and

$$B = \left\{ (X_1, X_2, X_3)^T \in \mathbb{E}^3 : X_1, X_2, X_3 \in (1, 3) \right\}.$$

Suppose

$$S = \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

and define

$$\phi(\mathbf{X}) = \mathbf{Y} + S(\mathbf{X} - \mathbf{Y}).$$

The map stretches the body by a factor of 2 in the first coordinate, contracts it by a factor of $\frac{1}{2}$ in the second, and leaves the third coordinate unchanged.



Example (A Stretch of a Rectangular Body, Cont.)

The coordinates of $\phi(\mathbf{X})$ are

$$x_1 = 2 + 2(X_1 - 2) = 2X_1 - 2,$$

$$x_2 = 2 + \frac{1}{2}(X_2 - 2) = 1 + \frac{1}{2}X_2,$$

$$x_3 = 2 + (X_3 - 2) = X_3.$$

Since $X_1, X_2, X_3 \in (1, 3)$, the image $B' = \phi(B)$ is

$$B' = \left\{ (x_1, x_2, x_3)^T \in \mathbb{E}^3 : x_1 \in (0, 4), x_2 \in \left(\frac{3}{2}, \frac{5}{2}\right), x_3 \in (1, 3) \right\}.$$



Result 4.2 (Rotation–Stretch Decompositions)

Let $\phi : B \rightarrow B'$ be a homogeneous deformation with a fixed point $\mathbf{Y}$, so that

$$\phi(\mathbf{X}) = \mathbf{Y} + F(\mathbf{X} - \mathbf{Y})$$

for some $F \in \mathcal{V}^2$. Let

$$F = RU = VR$$

be the right and left polar decompositions from Result 1.12. Then

$$\phi = \mathbf{r} \circ \mathbf{s}_1 = \mathbf{s}_2 \circ \mathbf{r},$$

where $\mathbf{r}$ is a rotation about $\mathbf{Y}$, while $\mathbf{s}_1$ and $\mathbf{s}_2$ are stretches from $\mathbf{Y}$.



Result 4.2 (Cont.)

The rotation and stretches are

$$\begin{aligned}\mathbf{r}(\mathbf{X}) &= \mathbf{Y} + R(\mathbf{X} - \mathbf{Y}), \\ \mathbf{s}_1(\mathbf{X}) &= \mathbf{Y} + U(\mathbf{X} - \mathbf{Y}), \\ \mathbf{s}_2(\mathbf{X}) &= \mathbf{Y} + V(\mathbf{X} - \mathbf{Y}).\end{aligned}$$

Proof.

Because R is a rotation tensor, $\mathbf{r}$ is a rotation about $\mathbf{Y}$. Because U and V are SPD, $\mathbf{s}_1$ and $\mathbf{s}_2$ are stretches from $\mathbf{Y}$.

For any $\mathbf{X} \in B$,

$$\begin{aligned}(\mathbf{r} \circ \mathbf{s}_1)(\mathbf{X}) &= \mathbf{r}(\mathbf{s}_1(\mathbf{X})) \\ &= \mathbf{Y} + R(\mathbf{s}_1(\mathbf{X}) - \mathbf{Y}).\end{aligned}$$



Proof (Cont.)

Using $\mathbf{s}_1(\mathbf{X}) = \mathbf{Y} + U(\mathbf{X} - \mathbf{Y})$ and $F = RU$,

$$\begin{aligned}(\mathbf{r} \circ \mathbf{s}_1)(\mathbf{X}) &= \mathbf{Y} + RU(\mathbf{X} - \mathbf{Y}) \\ &= \mathbf{Y} + F(\mathbf{X} - \mathbf{Y}) \\ &= \phi(\mathbf{X}).\end{aligned}$$

Similarly, using $\mathbf{r}(\mathbf{X}) = \mathbf{Y} + R(\mathbf{X} - \mathbf{Y})$ and $F = VR$,

$$\begin{aligned}(\mathbf{s}_2 \circ \mathbf{r})(\mathbf{X}) &= \mathbf{s}_2(\mathbf{r}(\mathbf{X})) \\ &= \mathbf{Y} + V(\mathbf{r}(\mathbf{X}) - \mathbf{Y}) \\ &= \mathbf{Y} + VR(\mathbf{X} - \mathbf{Y}) \\ &= \phi(\mathbf{X}).\end{aligned}$$

Therefore

$$\phi = \mathbf{r} \circ \mathbf{s}_1 = \mathbf{s}_2 \circ \mathbf{r}.$$





Extensions



Definition (Extension)

A homogeneous deformation ϕ is called an *extension* at the fixed point $\mathbf{Y}$ in the unit direction $\mathbf{e}$, $|\mathbf{e}| = 1$, if

$$\phi(\mathbf{X}) = \mathbf{Y} + F(\mathbf{X} - \mathbf{Y}),$$

where

$$F = I + (\lambda - 1)\mathbf{e} \otimes \mathbf{e}$$

for some $\lambda > 0$.

The parameter λ is the extension factor in the direction $\mathbf{e}$. Values $\lambda > 1$ give elongation, while $0 < \lambda < 1$ give contraction.

Action of an Extension



The tensor F extends every vector parallel to $\mathbf{e}$ by the factor λ . Indeed,

$$\begin{aligned} F\mathbf{e} &= \mathbf{e} + (\lambda - 1)(\mathbf{e} \cdot \mathbf{e})\mathbf{e} \\ &= \lambda\mathbf{e}. \end{aligned}$$

If $\mathbf{w} \cdot \mathbf{e} = 0$, then

$$F\mathbf{w} = \mathbf{w} + (\lambda - 1)(\mathbf{e} \cdot \mathbf{w})\mathbf{e} = \mathbf{w}.$$

Thus an extension changes only the component parallel to $\mathbf{e}$.



Proposition (An Extension is a Stretch)

Every extension deformation is a stretch deformation.

Proof.

It is enough to show that

$$F = I + (\lambda - 1)\mathbf{e} \otimes \mathbf{e}, \quad |\mathbf{e}| = 1, \quad \lambda > 0,$$

is symmetric positive definite. The identity and $\mathbf{e} \otimes \mathbf{e}$ are symmetric, so $F^T = F$. For any $\mathbf{x} \in \mathcal{V}$,

$$\begin{aligned} \mathbf{x} \cdot (F\mathbf{x}) &= \mathbf{x} \cdot (\mathbf{x} + (\lambda - 1)(\mathbf{e} \cdot \mathbf{x})\mathbf{e}) \\ &= |\mathbf{x}|^2 + (\lambda - 1)(\mathbf{e} \cdot \mathbf{x})^2 \\ &= |\mathbf{x}|^2 + \lambda(\mathbf{e} \cdot \mathbf{x})^2 - (\mathbf{e} \cdot \mathbf{x})^2. \end{aligned}$$



Proof (Cont.)

Suppose first that $\mathbf{x} \cdot \mathbf{e} \neq 0$. Decompose $\mathbf{x}$ into its component parallel to $\mathbf{e}$ and its component orthogonal to $\mathbf{e}$:

$$\mathbf{x}_e := (\mathbf{x} \cdot \mathbf{e})\mathbf{e}, \quad \mathbf{x}_s := \mathbf{x} - \mathbf{x}_e.$$

Since $\mathbf{x}_e \perp \mathbf{x}_s$, the Pythagorean theorem gives

$$\begin{aligned} |\mathbf{x}|^2 &= |\mathbf{x}_e|^2 + |\mathbf{x}_s|^2 \\ &\geq |\mathbf{x}_e|^2 \\ &= |(\mathbf{x} \cdot \mathbf{e})\mathbf{e}|^2 \\ &= (\mathbf{e} \cdot \mathbf{x})^2. \end{aligned}$$



Proof (Cont.)

Therefore, when $\mathbf{x} \cdot \mathbf{e} \neq 0$,

$$\begin{aligned}\mathbf{x} \cdot (F\mathbf{x}) &= |\mathbf{x}|^2 + \lambda(\mathbf{e} \cdot \mathbf{x})^2 - (\mathbf{e} \cdot \mathbf{x})^2 \\ &\geq \lambda(\mathbf{e} \cdot \mathbf{x})^2 \\ &> 0.\end{aligned}$$

On the other hand, if $\mathbf{x} \cdot \mathbf{e} = 0$ but $\mathbf{x} \neq \mathbf{0}$, then

$$\mathbf{x} \cdot (F\mathbf{x}) = |\mathbf{x}|^2 > 0.$$

Thus

$$\mathbf{x} \cdot (F\mathbf{x}) > 0 \quad \text{for every } \mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}.$$

Hence F is positive definite. Since F is also symmetric, it is SPD, and the extension is a stretch. □



Result 4.3 (Stretches as Extensions)

Let $\mathbf{s}_1$ and $\mathbf{s}_2$ be the stretches defined in Result 4.2, and let

$$\{\lambda_i, \mathbf{u}_i\}_{i=1}^3 \quad \text{and} \quad \{\lambda_i, \mathbf{v}_i\}_{i=1}^3$$

be eigensystems associated with U and V , respectively. Then

$$\mathbf{s}_1 = \mathbf{f}_1 \circ \mathbf{f}_2 \circ \mathbf{f}_3, \quad \mathbf{s}_2 = \mathbf{h}_1 \circ \mathbf{h}_2 \circ \mathbf{h}_3,$$

where $\mathbf{f}_i$ is an extension at $\mathbf{Y}$ by λ_i in the direction $\mathbf{u}_i$, and $\mathbf{h}_i$ is an extension at $\mathbf{Y}$ by λ_i in the direction $\mathbf{v}_i$.



Proof.

Recall from Result 4.2 that

$$\phi(\mathbf{X}) = \mathbf{Y} + F(\mathbf{X} - \mathbf{Y}), \quad F = RU = VR,$$

where

$$U = \sqrt{F^T F}, \quad V = \sqrt{F F^T}$$

are SPD tensors and R is a rotation. The tensors U and V have the same eigenvalues,

$$\lambda_1, \lambda_2, \lambda_3 > 0,$$

and the spectral decomposition theorem gives

$$U = \sum_{i=1}^3 \lambda_i \mathbf{u}_i \otimes \mathbf{u}_i, \quad V = \sum_{i=1}^3 \lambda_i \mathbf{v}_i \otimes \mathbf{v}_i,$$

for orthonormal eigenbases $\{\mathbf{u}_i\}$ and $\{\mathbf{v}_i\}$.



Proof (Cont.)

For $i = 1, 2, 3$, define

$$\mathbf{f}_i(\mathbf{X}) := \mathbf{Y} + F_i(\mathbf{X} - \mathbf{Y}),$$

where

$$F_i := I + (\lambda_i - 1)\mathbf{u}_i \otimes \mathbf{u}_i \quad (\text{no sum on } i).$$

Each $\mathbf{f}_i$ is an extension at $\mathbf{Y}$ by λ_i in the direction $\mathbf{u}_i$.

Similarly, define

$$\mathbf{h}_i(\mathbf{X}) := \mathbf{Y} + H_i(\mathbf{X} - \mathbf{Y}), \quad H_i := I + (\lambda_i - 1)\mathbf{v}_i \otimes \mathbf{v}_i,$$

with no summation on i . Each $\mathbf{h}_i$ is an extension in the direction $\mathbf{v}_i$.



Proof (Cont.)

The composition of the second and third extensions satisfies

$$\begin{aligned}(\mathbf{f}_2 \circ \mathbf{f}_3)(\mathbf{X}) &= \mathbf{f}_2(\mathbf{f}_3(\mathbf{X})) \\ &= \mathbf{Y} + F_2(\mathbf{f}_3(\mathbf{X}) - \mathbf{Y}) \\ &= \mathbf{Y} + F_2 F_3(\mathbf{X} - \mathbf{Y}).\end{aligned}$$

Consequently,

$$\begin{aligned}(\mathbf{f}_1 \circ \mathbf{f}_2 \circ \mathbf{f}_3)(\mathbf{X}) &= \mathbf{f}_1((\mathbf{f}_2 \circ \mathbf{f}_3)(\mathbf{X})) \\ &= \mathbf{Y} + F_1 F_2 F_3(\mathbf{X} - \mathbf{Y}).\end{aligned}$$

Thus it remains to show that $F_1 F_2 F_3 = U$.



Proof (Cont.)

Because $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis,

$$(\mathbf{u}_i \otimes \mathbf{u}_i)(\mathbf{u}_j \otimes \mathbf{u}_j) = (\mathbf{u}_i \cdot \mathbf{u}_j)\mathbf{u}_i \otimes \mathbf{u}_j = 0 \quad \text{when } i \neq j.$$

Therefore the cross terms vanish, and

$$\begin{aligned} F_1 F_2 F_3 &= \prod_{i=1}^3 [I + (\lambda_i - 1)\mathbf{u}_i \otimes \mathbf{u}_i] \\ &= I + \sum_{i=1}^3 (\lambda_i - 1)\mathbf{u}_i \otimes \mathbf{u}_i. \end{aligned}$$

Here i is a dummy index in the sum.



Proof (Cont.)

The identity tensor has the representation

$$I = \sum_{i=1}^3 \mathbf{u}_i \otimes \mathbf{u}_i.$$

Hence

$$\begin{aligned} F_1 F_2 F_3 &= \sum_{i=1}^3 \mathbf{u}_i \otimes \mathbf{u}_i + \sum_{i=1}^3 (\lambda_i - 1) \mathbf{u}_i \otimes \mathbf{u}_i \\ &= \sum_{i=1}^3 \lambda_i \mathbf{u}_i \otimes \mathbf{u}_i \\ &= U. \end{aligned}$$



Proof (Cont.)

We already showed that

$$(\mathbf{f}_1 \circ \mathbf{f}_2 \circ \mathbf{f}_3)(\mathbf{X}) = \mathbf{Y} + F_1 F_2 F_3 (\mathbf{X} - \mathbf{Y}).$$

Therefore

$$(\mathbf{f}_1 \circ \mathbf{f}_2 \circ \mathbf{f}_3)(\mathbf{X}) = \mathbf{Y} + U(\mathbf{X} - \mathbf{Y}) = \mathbf{s}_1(\mathbf{X}).$$



Proof (Cont.)

The same argument in the orthonormal eigenbasis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ of V gives

$$H_1 H_2 H_3 = \sum_{i=1}^3 \lambda_i \mathbf{v}_i \otimes \mathbf{v}_i = V.$$

Also,

$$(\mathbf{h}_1 \circ \mathbf{h}_2 \circ \mathbf{h}_3)(\mathbf{X}) = \mathbf{Y} + H_1 H_2 H_3 (\mathbf{X} - \mathbf{Y}).$$

It follows that

$$(\mathbf{h}_1 \circ \mathbf{h}_2 \circ \mathbf{h}_3)(\mathbf{X}) = \mathbf{Y} + V(\mathbf{X} - \mathbf{Y}) = \mathbf{s}_2(\mathbf{X}).$$

Thus

$$\mathbf{s}_1 = \mathbf{f}_1 \circ \mathbf{f}_2 \circ \mathbf{f}_3, \quad \mathbf{s}_2 = \mathbf{h}_1 \circ \mathbf{h}_2 \circ \mathbf{h}_3.$$

